

SUBODH UNIYAL

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EDUCATION

Bachelor of Technology in Computer Science

Indian Institute of Information Technology Kottayam (IIITK), Kerala

Dec 2021 - Present

CGPA : 8.85

Higher Secondary Education

Shri Digamber Jain Sr. Sec. School, Sikar, Rajasthan

April 2018 - March 2020

Percentage: 85.8

WORK EXPERIENCE

Machine Learning (LLM) Intern, *capSwipe*

Dec 2023 - Present

- Conducted research on **State-of-the-art Text-to-SQL models and techniques**, implementing Pipelines for utilizing **LLMs for NL-to-SQL tasks**, experimenting with over **10+ LLMs**.
- Integrated and prompt engineered LLMs for advanced **Information retrieval** and Generation system, resulting in up to a **10% performance boost**.

Subject Matter Expert - AI/ML, *House of couton private limited*

Sept 2023 - Nov 2023

- Generated comprehensive reports on **AI/ML findings** to facilitate decision-making.
- Conducted daily AI/ML research, including coding and evaluation for publication based on **peer-reviewed papers**.

Research Intern, *IIIT Kottayam*

June 2023 - Present

- Presented comprehensive findings, methodology, and implemented machine learning algorithms for **research papers and over 10 assignments**, extending the research over **Text, Image and Multi-modality**.
- Utilized advanced techniques like **SHAP, LIME, Anchor, and IG** values to enhance AI model explainability, resulting in a **20% decrease in misclassifications** and increased transparency..

TECHNICAL SKILLS

Languages/Frameworks: Python, SQL, C++, HTML/CSS, JavaScript, Tensorflow, Flask, Django, Keras, Scikit-Learn, OpenCV, Huggingface, Pandas, Numpy, Matplotlib, Seaborn, Pytorch, Langchain, LlamaIndex

Tools/Databases: Git, Linux, REST Api, PowerBI, Docker, MySQL, MongoDB, Firestore, PostgreSQL, AWS

PROJECTS

The Wise Saint: An AI Tutor

June 2024

An AI tutor providing detailed mathematical solutions with reasoning and graph-based explanations, leveraging **Langchain**, **Langgraph** and the **Gemini API**. Features a multi-modality support and CI/CD via **GitHub Workflows** and **AWS**.

- **Advanced Multi-modality Support:** Utilizes a **RAG system** that seamlessly integrates text and image data to deliver mathematical solutions, achieving an **accuracy rate of over 90%** with support of **10+ Knowledge Bases**.
- **Graph and Plot-Based Explanations:** Provides detailed explanations with visual support using graphs and plots to enhance understanding, capable of generating **with less than 2 seconds of generation time**.

Career Counselling Chatbot

February 2024

A cutting-edge career counselling chatbot employing advanced language models and the Retrieval-Augmented Generation (RAG) framework to provide personalized career guidance to users with context to over **1100+ career paths**.

- **Large Model Support:** RAG model incorporated based on **Google Gemma**, and tested on **Zephyr-7b, LLama-2**, and other similar Billion parameters models, with support for **4-bit quantization**.
- **Efficient and Responsive:** By utilizing frameworks like **LlamaIndex, Langchain** and **Transformers**, delivering personalized recommendations with an average response time of **less than 5 seconds**.

Sentify: Real-time Sentiment Analysis

December 2023

Sentify focuses on providing real-time sentiment analysis of social media platforms like **Twitter** and **YouTube**. Leveraging **advanced NLP and audio processing**, capable of **analyzing text and audio content** to develop actionable insights.

- **Model Integration:** Integrates advanced Large models including **Roberta, Hubert-large, DistilBERT Multilingual**, and others with total of **5+ models**, resulting in a robust sentiment analysis system achieving over **97% accuracy**.
- **Multilingual and Efficient:** Sentify, a cutting-edge platform supporting **104 languages**, **swiftly analyzing 1000+ posts in seconds**; revolutionized sentiment analysis, ensuring speed and scalable operations.

CERTIFICATIONS & ACHIEVEMENTS

- Selected for **publication** at the **International Conference on Data Science & Exploration in Artificial Intelligence, 2024**, results demonstrated biasness and measured responsibility of the Legal AI over **6 Metrics**.
- **IBM Data Science Professional Certificate** and **Generative AI using Large Language Models** by DeepLearning.AI
- **Grand Finale contender at Smart India Hackathon 2023**, showcasing exceptional innovation and analytical acumen, and attended an **IDE bootcamp** hosted by **AICTE**, acquiring vital skills in **business development and innovation**.
- Led the **AI and Data Science (AI&DS) Club at IIIT Kottayam**, overseeing project research and development with a team of SME's to complete **6 major projects**.